NASA SPACE APPS CHALLENGE 2026

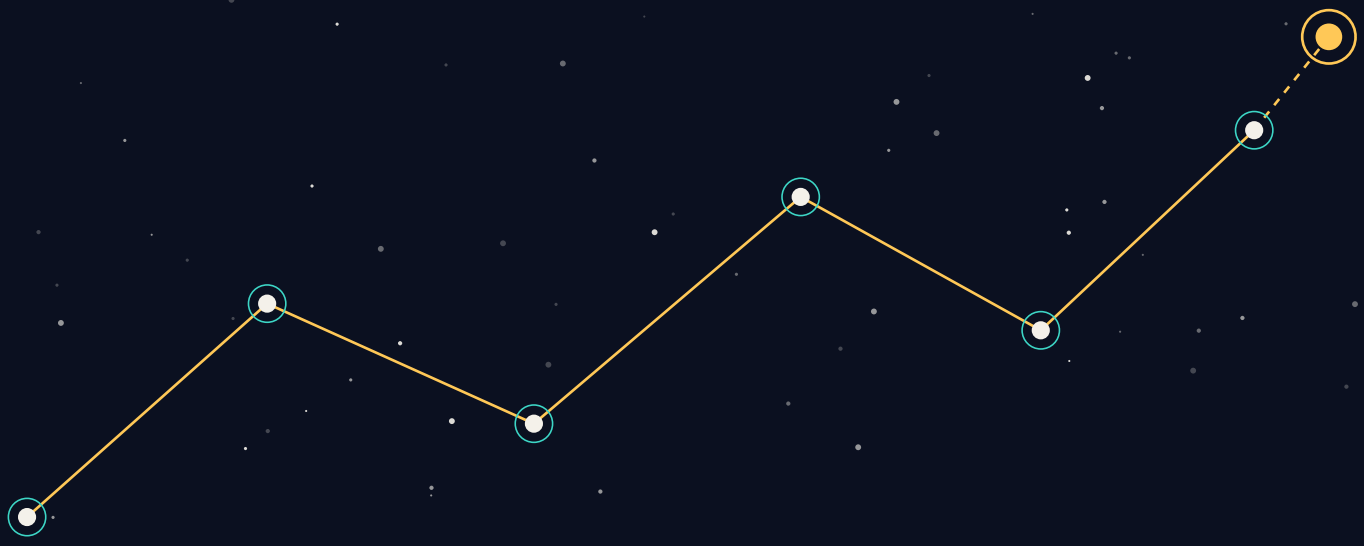
BUILT FOR A QUESTION

Developer and presentation brief

Challenge: Abandoned but not Forgotten: Storytelling about NASA's Discarded Equipment on the Moon and Mars

THE QUESTION OF THE MOMENT

How far away is the Moon, exactly?



Every machine NASA left behind was built to answer one question.

Each answer created the next question. The website is the journey through them, and the last question belongs to the visitor.

Version 1.0 | October 30, 2026 | Audience of this document: developers, designers, video team

Concept status: LOCKED | Not an official NASA product. NASA does not endorse this project.

HOW TO USE THIS BRIEF

Read this first

This document is the single source of truth for what we are building. A developer who reads it from start to finish should be able to build the whole website without asking what the story is, what each screen does, or where each fact comes from. The video team uses sections 1, 2 and 13.

Tags used throughout

Tag	Meaning
[V]	Verified during research for this brief. The source is named in section 12.
[C]	Confirm on a NASA page before it appears on screen. Written from general knowledge, not checked line by line.
MUST	Needed for the project to make sense. Build first.
SHOULD	Strongly improves the project. Build if MUST items are done.
COULD	Nice to have. Cut without regret.

LOCKED DECISIONS (do not change without team agreement)
L1 It is a story website, not a game. No score, no timer, no win or lose.
L2 The story is one scroll-driven journey outward from Earth: 6 chapters, then a finale.
L3 The signature feature is the **Question Banner** that rewrites itself between chapters.
L4 Every chapter follows the same 4 steps: Meet it, How it works, What we learned, Where it is now.
L5 Hardware and engineering are the heroes. Each part of each machine has a job and a lesson.
L6 Every sentence carries a Truth Label (FACT, INFERRED or IMAGINED) or is plainly factual with a source.
L7 The chat is a small "Ask the engineer" box that answers only from our fact cards.
L8 The finale lets the visitor write a question and download a Constellation card. Nothing is stored on a server.

What developers may change: library choices, file structure, exact animation timings (within the ranges given), small layout details, which extra machines to add if time allows. **What developers may not change:** anything under LOCKED DECISIONS, the chapter order, the question chain, or the wording of the six main questions.

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WHY THIS EXISTS

1 Project summary and judging fit

One sentence. A scroll-driven journey from Earth to interstellar space in which every machine NASA left behind is shown through the question it was built to answer, the clever engineering that let it answer, and what we learned, with a real photo of where it sits today.

The problem. Young people hear about active missions. The machines left on the Moon, on Mars and in deep space are usually presented as a list of old objects. Lists do not create feeling, and they hide the real story: each machine passed a question to the next one.

Our answer. We do not present machines as a gallery. We present them as a relay. The Question Banner at the top of the screen always shows the question that the current machine was built to answer. When a chapter ends, the banner rewrites itself into the next machine's question and the camera flies on. At the end the banner goes blank and asks the visitor for theirs.

Audience. School-age space fans, ages 9 to 14. Secondary audience: teachers and parents, and the Space Apps judges.

Challenge brief fit.

The brief asks for	How we answer
A narrative focus with real facts and mission data	One story spine (the question chain). Every chapter is built from a source-linked fact card.
Show how missions operated, not just list facts	Tap the parts of a 3D model. Each part says "my job" and "what we learned because of me".
Connect the engineering to the science	Chapter step 2 (engineering) leads directly into step 3 (discovery). They are never separated.
Make people feel a connection to the machines	Status chips (Still talking, Silent, Retired), honest "last signal" moments, and the hand-off question.
Accuracy matters	Truth Labels, a source drawer on every chapter, and a chat that cannot go beyond the fact cards.
Accessible format	Keyboard navigation, reduced motion, text alternatives, read aloud, and image fallbacks if 3D fails.

Judging fit. Past Space Apps guides score projects on creativity, scientific validity, relevance to the challenge, presentation and impact. The 2026 judging guide is released on November 13. Re-read it and adjust the video and project page to match.

Criterion	What judges will look for	Where we deliver it
Creativity	A fresh idea, not another gallery	Question Banner rewrite, relay chain, Constellation finale
Validity	Correct science and honest sourcing	Fact cards, [V]/[C] review, Truth Labels, refusal-safe chat
Relevance	Clearly answers this challenge	Left-behind hardware on the Moon, Mars and deep space, with discoveries
Presentation	A clear, memorable demo and video	Section 13: four unforgettable moments, 30-second cut
Impact	Value for learners	Ages 9 to 14, teacher-friendly facts, share card, accessible

Timely hook

NASA's projected date for Voyager 1 to reach one light-day from Earth is **November 18, 2026**, about 10:16 UTC [V]. That is three days after submissions close, while judging begins. The final chapter shows the live distance counter, so the project is "alive" while judges look at it.

THE BIG IDEA

2 The concept and the question chain

Core idea. No machine out there was abandoned. Each one finished a job and left a question behind for the next one. Following the questions, a visitor travels from the Moon to Mars to the edge of the Sun's reach, and ends by adding a question of their own.

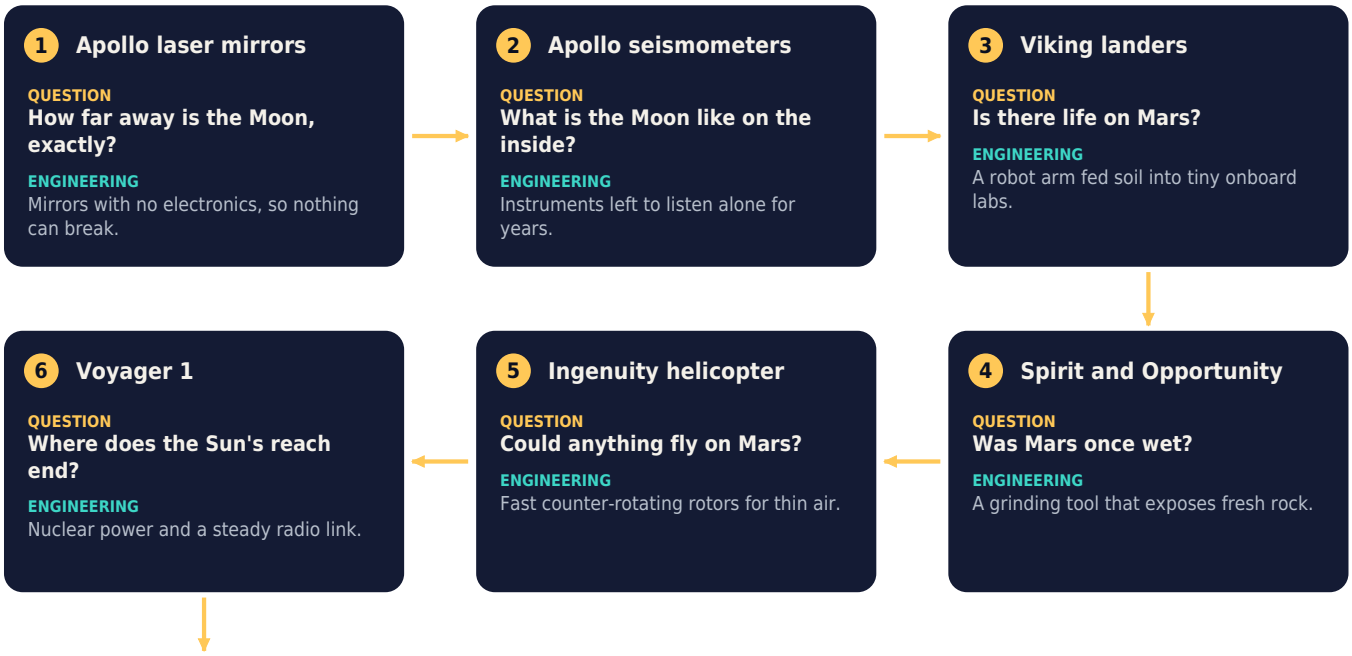


Figure 2.1 The relay: six machines, six questions, one engineering idea each. Read the cards left to right on the top row, then right to left on the bottom row.

The Question Banner (signature feature). A fixed banner at the top of every screen shows the question of the moment, in Question Gold on a dark panel. It is the spine of the website:

- It is the first thing visitors read on the landing screen and the last thing they write in the finale.
- Between chapters it **rewrites itself**: the old question erases right to left (600 ms), the new one types in at 35 ms per character, and the gold underline pulses once.
- It is always readable by screen readers (aria-live="polite") and respects reduced motion (a 300 ms crossfade).
- It must never be hidden, covered, or shortened on mobile. If a question wraps to two lines, the banner grows.

The hand-off lines. Between chapters the narrator says one line that connects the old answer to the new question. These are fixed copy:

From	To	Hand-off line (shown above the banner as it rewrites)
Ch 1	Ch 2	The Moon's distance: measured. But what is inside it?
Ch 2	Ch 3	The Moon turned out to be quiet, layered rock. What about the world next door?
Ch 3	Ch 4	The Viking labs left a puzzle. To solve it, scientists needed a better clue: water.
Ch 4	Ch 5	Mars once had water. Now, how do we explore it faster and farther?
Ch 5	Ch 6	A helicopter proved machines can go new places. How far can they really go?
Ch 6	Finale	Voyager is still going. Your turn.

What we are deliberately NOT building

- A game with a score, levels or a timer. (A friend's idea, "pick Moon or Mars and find rovers in 3D", is a search task with no story. We build the story first.)
- A museum grid where visitors click any machine in any order. Order matters here: the chain is the story.
- A free-roam 3D world. 3D is used inside chapters, only to show how a machine works.
- A general-purpose AI chatbot. The chat can only answer from our fact cards.
- Invented quotes from machines. If a machine "speaks", the line is labeled IMAGINED and is never presented as a real transmission.

WORDS AND TRUST

3 Audience, voice and Truth Labels

Rule	Specification
Age and reading level	Ages 9 to 14. Write at roughly US grade 5 to 7. Check with a readability tool; aim below grade 8.
Sentence length	Average under 14 words. Never more than 22.
Block length	At most 70 words per step. One idea per step.
Voice	Curious, warm, precise. Explain like a good teacher, not like a textbook. No baby talk.
Person	Narrator speaks to "you". Machines are "it", never "he" or "she".
Numbers	Give one precise number and one comparison a child can picture. Example: "about 17 km every second, faster than any car or plane".
Jargon	Each technical word is explained in the sentence where it first appears. Optional tap-to-define on key words (COULD).
Emotion	Allowed and encouraged, but only from true events: a long mission, a quiet ending, a handed-off question.
Do not	Do not say "dead" for machines in headings. Do not claim a machine "felt" anything as fact. Do not use sarcasm or fear.

Truth Labels. A small label can be attached to any sentence block. It teaches children how science communication works and protects the project from errors.

FACT INFERRED IMAGINED STILL TALKING STILL ANSWERING SILENT RETIRED

First three: Truth Labels (on text). Last four: Status Chips (on each machine). STILL ANSWERING = a passive device that replies when contacted.

Label	Use it when	Example
FACT	Measured or documented, with a source on the fact card.	"Voyager 1 sends data at about 160 bits per second." [V]
INFERRED	Scientists' interpretation of data. A good explanation, not a direct observation.	"Minerals like these usually form in water, so Mars was probably wet here." [C]
IMAGINED	Dramatization we wrote to help feel the moment. Always labeled, kept rare (maximum one per chapter).	"Imagine the quiet after the last signal."

Footer disclaimer (on every page of the site): "Dramatized voice, facts from NASA. This is a student project for the NASA Space Apps Challenge. It is not an official NASA website and is not endorsed by NASA."

Accuracy rule for writers
A claim goes on the site only if the fact card has (1) a source link, (2) the date the team checked it, and (3) a second team member's initials. A claim tagged [C] in this brief cannot ship until all three exist.

SCREENS AND GLOBAL UI

4 Experience architecture

The site is one long, scroll-driven page, split into screens. Each chapter is a "pinned" section: the scene stays in view while the visitor moves through four steps. A visitor can also use the Next and Back buttons or the keyboard (Space, Arrow keys, Page Down).



Figure 4.1 Sitemap. The path S0 to S8 is the whole story. S9 is reachable from the menu at any time.

Screen list

ID	Screen	Purpose	Banner shows
S0	Landing and prologue	Hook. Explain the idea in 2 lines. One button: "Begin".	How far away is the Moon, exactly?
S1	Chapter 1: Laser mirrors	Moon distance, mirrors that need no power	How far away is the Moon, exactly?
S2	Chapter 2: Seismometers	Inside of the Moon, instruments left alone	What is the Moon like on the inside?
S3	Chapter 3: Viking landers	Search for life; a puzzle, not a win	Is there life on Mars?
S4	Chapter 4: Spirit and Opportunity	Water on Mars; long missions; silent endings	Was Mars once wet?
S5	Chapter 5: Ingenuity	First flights on another planet	Could anything fly on Mars?
S6	Chapter 6: Voyager 1	Still talking. Live distance. The open question	Where does the Sun's reach end?
S7	Finale: your question	Banner goes blank; visitor writes a question	(empty, cursor blinking)
S8	Constellation card	All 7 questions as a star map. Download or share.	What will you ask next?
S9	Sources and About	Every source, credits, licenses, disclaimer, team	(hidden)

4.2 Global UI components

These stay on screen across the whole journey.

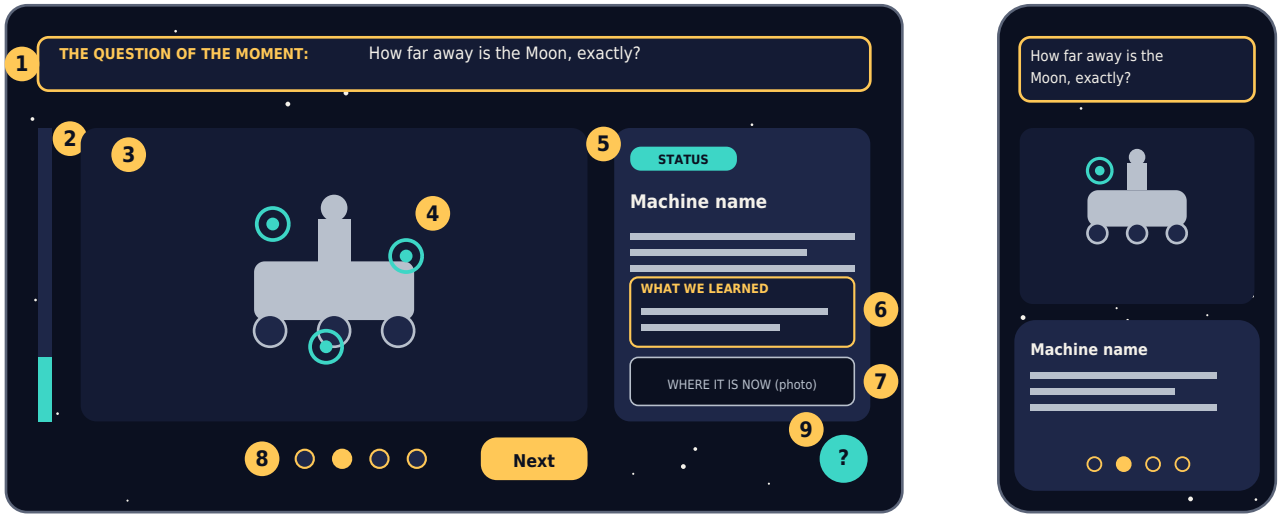
Component	Behavior and spec	Priority
Question Banner	Fixed at the top. Full width, 56 px tall on desktop, grows if text wraps. Question Gold text on Midnight Panel with a 2 px gold underline. Rewrites between chapters (see section 2). aria-live="polite".	MUST
Distance Ruler	A thin vertical bar on the left edge (desktop) or a thin horizontal bar under the banner (mobile). Log scale from the Moon to Voyager 1 (Figure 5.3). A teal marker moves as the visitor moves through chapters. Tooltip on hover: "You are about X km from Earth".	SHOULD
Progress dots	Six chapter dots plus a finale dot. Current dot is gold. Clicking a dot jumps back to a chapter already visited. Future dots are not clickable until visited (the order matters).	MUST
Step dots	Four small dots inside a chapter (Meet it, How it works, What we learned, Where it is now) with Back and Next buttons.	MUST
Menu	Icon at top right. Opens a drawer: chapter list (visited ones), Sources and About, Settings.	MUST
Settings	Reduce motion (default follows OS), text size (A, A+, A++), read aloud on/off, high contrast, language EN/BN (COULD).	MUST (motion, text size)
Ask the engineer	Floating round teal button bottom right. Opens a small chat panel (section 8).	SHOULD
Sources drawer	Button "Sources" inside every chapter. Lists the fact-card sources used on that screen with links.	MUST
Status chip	On each machine: STILL TALKING (teal), SILENT (rust), RETIRED (silver). Includes a date of last contact where known [C].	MUST

ONE TEMPLATE, SIX TIMES

5 The chapter template, camera and animation

All six chapters use the **same template**. Build it once, then fill it with content from a fact card. This is the single most important engineering decision: it keeps the project finishable in 48 hours and makes every chapter feel part of one story.

5.1 Anatomy of a chapter screen



DESKTOP (16:9) MOBILE (portrait, bottom sheet)

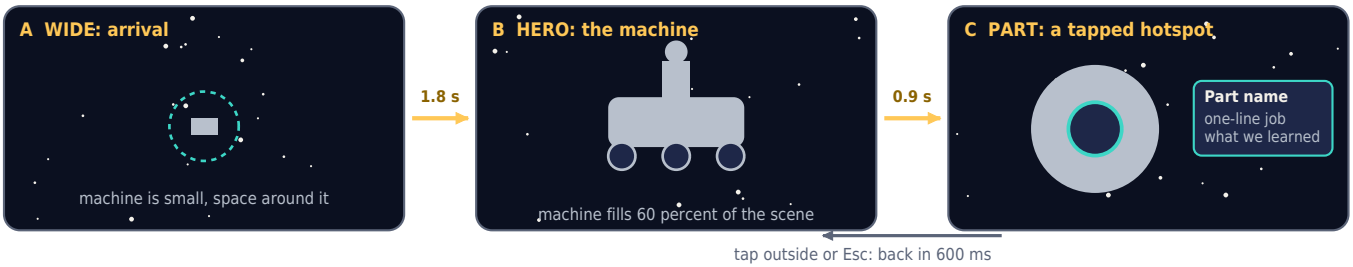
Figure 5.1 Chapter screen wireframe. Numbers match the table below. Dark background is the real site background.

#	Region	Specification
1	Question Banner	See 4.2. Shows this chapter's question during steps 1 to 4.
2	Distance Ruler	Marker shows how far from Earth this machine is (Figure 5.3).
3	Scene (3D viewer)	60 percent of width on desktop, top 45 percent of the screen on mobile. Dark gradient background. Soft key light plus a rim light. Slow auto-rotate when idle. Falls back to a still PNG if WebGL fails.
4	Hotspots	3 or 4 pulsing teal rings on the model. Tap or press Enter to open. Each has an accessible name ("Corner mirrors, button").
5	Story panel	Status chip, machine name, current step copy. Right column on desktop, bottom sheet on mobile (peek 30 percent, expandable).
6	What we learned	Gold-outlined box inside the panel at step 3. Contains the discovery and a Truth Label.
7	Where it is now	Real orbiter or NASA photo at step 4, with caption and credit. Replaces the scene via crossfade.
8	Step dots, Back, Next	Four steps. Next is never disabled. On the last step Next reads "Next question".
9	Ask the engineer	Opens the chat panel scoped to this chapter's fact card first, then all cards.

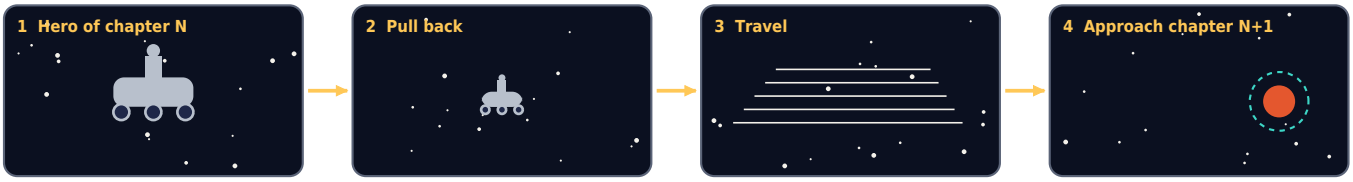
5.2 The four steps

Step	Name	Camera	What the visitor sees and does
1	Meet it	A (wide) to B (hero), 1800 ms	Status chip, name, 2 sentences. Hotspots hidden. Banner already shows the question. One short "last signal" or "still talking" line.
2	How it works	B; tap a hotspot to go to C, 900 ms	Hotspots pulse. Tapping opens the part card: its job, then what we learned because of it. Opening 2 or more parts makes the Next button glow. Never block Next.
3	What we learned	B, scene dims 30 percent	The part that made the discovery possible stays highlighted. The "What we learned" box appears with a Truth Label and one key number.
4	Where it is now	Crossfade to photo, 600 ms	Real photo, caption, credit, Sources button. Next triggers the hand-off to the next chapter (5.5).

5.3 Camera and zoom behavior



BETWEEN CHAPTERS (the hand-off): zoom out, travel, zoom in



During step 2 to 3 the banner rewrites itself. Total hand-off about 4.6 seconds. With reduced motion: a 300 ms crossfade instead.

Figure 5.2 Camera states (top) and the chapter-to-chapter hand-off (bottom). All times are animation durations in milliseconds.

Item	Specification
Camera	Perspective camera, 35 degree field of view. Distance from the model: A = 6 times the model radius, B = 2.6 times, C = 1.1 times the part's radius.
Rotate	Desktop: drag to rotate in state B, limited to 35 degrees left or right and 15 degrees up or down. Mobile: one finger scrolls the page; a "Rotate" toggle button enables one-finger rotation. Provide a Reset view button.
Zoom	No pinch or wheel zoom inside the scene (it fights with page scroll). Zoom happens only through hotspots (B to C) and the hand-off.
Auto-rotate	0.1 radians per second when idle for 3 seconds. Off with reduced motion.
Focus	When a hotspot opens, keyboard focus moves to the part card. Esc closes it and returns focus to the hotspot.

5.4 Distance Ruler

Log scale: each tick is 10 times farther than the one before. This is the UI Distance Ruler (see 5.4).

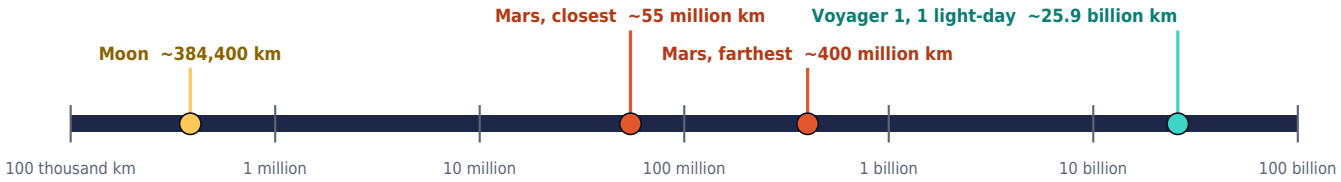


Figure 5.3 The Distance Ruler on a log scale. Marker positions use the formula x = log10(distance in km). Mars distance varies; use the range.

The marker sits at the Moon for chapters 1 and 2, inside the Mars range for chapters 3 to 5, and at Voyager 1 for chapter 6. Values shown are approximate and tagged [C], except Voyager 1's one-light-day figure, which comes from NASA's announcement [V].

5.5 Animation specification

Animation	Duration	Easing	Reduced-motion version
Banner old question erases (right to left)	600 ms total	linear	Crossfade 300 ms
Banner new question types in	35 ms per character, max 2.6 s	steps	Appears with the crossfade
Banner gold underline pulse	500 ms, once	ease-out	No pulse
Camera A to B (arrival)	1800 ms	easeOutCubic	Jump cut + 300 ms fade
Camera B to C (tap a part)	900 ms	easeInOutCubic	300 ms fade
Camera C to B (close part)	600 ms	easeInOutCubic	300 ms fade
Step text change	250 ms fade + 12 px rise	ease-out	Fade only
Hotspot ring pulse	1.6 s loop	ease-in-out	Static ring, no loop
3D to photo crossfade	600 ms	ease	300 ms
Hand-off: pull back	1200 ms	easeInOut	Skipped
Hand-off: travel (star streaks, ruler marker slides)	1600 ms	linear	Ruler marker jumps; no streaks
Hand-off: approach next chapter	1800 ms	easeOutCubic	300 ms fade
Starfield parallax (3 layers)	follows scroll	-	Static starfield

Safety rule
Nothing may flash more than 3 times per second. No full-screen white flashes. Autoplaying sound is forbidden; any sound is opt-in with captions.

5.6 Scroll and navigation rules

- Use native scrolling. Each chapter is a tall section (400 vh) with a sticky 100 vh stage inside it. Do not hijack the wheel or touch events.
- Next and Back buttons scroll smoothly to the next step (instant with reduced motion). Keyboard: Space or Page Down = Next, Shift+Space or Page Up = Back, Esc = close part card.
- Deep links use a plain hash only: #ch1 to #ch6, #finale, #sources. Do not put question text or any visitor data in the URL.
- The page title updates per chapter (for history and screen readers). Focus moves to the chapter heading when a chapter starts.
- If the visitor scrolls fast, skip the hand-off animation but still rewrite the banner.

CONTENT FOR THE TEMPLATE

6 Chapter-by-chapter specification

Each chapter below is a fill-in for the template in section 5. Copy is draft text at the right reading level. Developers: put copy in the JSON fact cards (section 8.3), not in the components. Writers: do not change a [V] claim without checking the source; do not publish a [C] claim before it is confirmed.

Models and photos
NASA 3D Resources has models for many missions, but we must confirm for each machine that a model exists and what its credit line is. If none exists, build a simple, honest 3D illustration in three.js and label it "Illustration". Never present an illustration as an official NASA model. All photos must keep their credit line (for example "NASA/JPL-Caltech/UArizona").

6.1 Chapter 1: Apollo laser mirrors

Banner in	**How far away is the Moon, exactly?**
Banner out (next chapter)	**What is the Moon like on the inside?**
Status chip	STILL ANSWERING [C]
Where it is	Moon. Apollo 11, 14 and 15 sites [C]
3D model file	ch1-retroreflector.glb (or illustration: a flat panel with an array of small corner-cube circles)
"Where it is now" photo	ch1-lroc-apollo15.jpg from LROC featured sites (Apollo 15 close-up shows the lander base and other hardware [V])

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Corner mirrors	Each tiny mirror is shaped like the corner of a cube. Light that hits it bounces straight back the way it came.	Light returns to Earth even if the panel is not aimed perfectly. [C]
2	The panel	Many mirrors sit in one frame. More mirrors send more light back.	Very little light returns, so scientists count the few light particles that do. [C]
3	Laser on Earth (marker, not on the model)	A telescope fires a short laser pulse. An exact clock times the trip there and back.	Light moves at a known speed, so time becomes distance.
4	Nothing electric	There is no battery, wire or computer in it.	A simple design with nothing to wear out can last for decades. [C]

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	Apollo astronauts left panels of tiny mirrors on the Moon. No batteries. No wires. No computer. Just one job: send light straight back.	FACT [C]
2 How it works	Tap the glowing dots. A telescope on Earth fires a laser at the panel. The mirrors send it right back. A clock times the trip.	FACT
3 What we learned	The round trip takes about two and a half seconds [C]. Because we know how fast light travels, the time gives the distance to within centimeters [C]. After decades of measuring, scientists found the Moon is drifting away from Earth by about 3.8 centimeters a year [C].	FACT [C]
4 Where it is now	These mirrors can still answer lasers today [C]. The panel is too small to pick out from orbit, but the site is on the map. [C]	FACT [C]

Facts used and how to verify them

Claim	Tag	Where to verify
Mirror panels left by Apollo 11, 14 and 15	C	NASA LRO pages; NSSDCA entries for each mission
Round trip about 2.5 s (average distance 384,400 km, light 299,792 km/s: 2 x 384,400 / 299,792 = 2.56 s)	C	Check average distance on a NASA Moon fact sheet
Moon drifts away about 3.8 cm per year	C	NASA or an observatory lunar-ranging page
Still in scientific use	C	A current lunar laser ranging station page

Developer notes

- SHOULD: "Fire laser" mini-interaction. A pulse crosses from Earth to the panel and back in 2.56 seconds real time with a stopwatch. Caption: "This is real speed."
- The panel is on the Moon but the laser source is not. Draw the laser from outside the frame to avoid confusion.

6.2 Chapter 2: Apollo seismometers

Banner in	**What is the Moon like on the inside?**
Banner out (next chapter)	**Is there life on Mars?**
Status chip	RETIRED [C]
Where it is	Moon. Apollo 12 to 17 stations [C]. (Apollo 11 carried a simpler version [V])
3D model file	ch2-alsep.glb (or illustration of a science station: box, antenna, small instrument)
"Where it is now" photo	ch2-lroc-apollo17.jpg Apollo 17 site: the cross-shaped path where astronauts set up listening instruments is visible from orbit [V]. Alternate: Apollo 12 image with the bright L-shape marking cables [V]

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Seismometer	It feels tiny shakes in the Moon's ground.	Moonquakes exist, and how shaking travels reveals layers inside. [C]
2	Central station	The box with the radio, clock and electronics. It sent the measurements to Earth.	Data kept arriving for years. [V]
3	Power unit	Later stations used a nuclear power unit that makes electricity from heat. [C]	It kept working through the cold lunar night, when sunlight is not available. [C]
4	Antenna	It pointed at Earth to send the data home.	Even the best sensor is silent without a way to talk.

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	After the astronauts flew home, the science stations they left behind kept working. They sent data to Earth for years, with nobody there to watch.	FACT [V]
2 How it works	Tap the glowing dots. A seismometer feels the Moon's ground shake. The central station turns the shaking into radio signals. The antenna sends them home.	FACT
3 What we learned	The Moon has quakes of its own. Scientists studied how the shaking travels and worked out that the Moon has layers inside, like Earth: a crust, a mantle and a core. [C]	INFERRED
4 Where it is now	The stations are retired [C], but the recordings were not lost. Data recovered from old archive tapes has been published for scientists to use today. [V]	FACT [V]

Facts used and how to verify them

Claim	Tag	Where to verify
Stations on Apollo 12 to 17; Apollo 11 had a simpler version	C / V	LRO fact sheet [V] says Apollo 11 carried a simpler science package; station list: NSSDCA
Data recovered from archival tapes and published through the PDS	V	PDS Geosciences Node, Apollo ALSEP data page
Apollo 17 cross-shaped path where listening instruments were set up is visible from orbit	V	NASA Goddard / spaceflightnow caption of the LRO Apollo 17 image
Stations shut down in 1977	C	NSSDCA Apollo ALSEP page
Layers: crust, mantle, core	C	NASA Moon science page

Developer notes

- SHOULD: tapping the seismometer makes the model vibrate gently. Optional low rumble: off by default, with a caption "[low rumble]".
- Do not claim how many moonquakes were recorded unless the number is taken from a NASA page and tagged [V].

6.3 Chapter 3: Viking landers

Banner in	**Is there life on Mars?**
Banner out (next chapter)	**Was Mars once wet?**
Status chip	SILENT [C]
Where it is	Mars. Two landers, landed in 1976 [C]
3D model file	ch3-viking.glb (NASA 3D Resources if available [C])
"Where it is now" photo	ch3-hirise-viking1.jpg HiRISE view of the Viking 1 site: lander, back shell about 260 m away, heat shield farther [V]; rock nicknamed "Big Joe" near Viking 1 [V]

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Sampler arm	A robot arm scooped soil and dropped it into the lander's labs. [C]	A robot can dig, deliver and test with no human hand.
2	Mini labs	Small instruments tested the soil for signs of living things. [C]	The soil was chemically surprising, but that alone did not prove life. [C]
3	Cameras	They photographed the ground and sky around the lander. [C]	Pictures helped the team choose where the arm should dig.
4	Antenna	It sent data to Earth, directly or through orbiters circling above. [C]	A lander and an orbiter working as a team. [C]

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	In 1976, two Viking landers touched down on Mars. Each carried a robot arm and tiny labs. Their job was huge: look for life. [C]	FACT [C]
2 How it works	Tap the glowing dots. The arm scoops soil. The labs test it. The cameras watch. The antenna sends the news home.	FACT [C]
3 What we learned	The labs gave puzzling results. Most scientists concluded they did not show life, though a few still debate it. [C] No organic molecules were found at the places tested. [C] The answer was not yes or no. It was: we need a better clue. Follow the water.	INFERRED
4 Where it is now	Viking 1 stopped talking in November 1982 [C]. From orbit we can still see the lander, its back shell, and rocks the lander photographed in 1976, like one nicknamed "Big Joe". [V]	FACT [V]

Facts used and how to verify them

Claim	Tag	Where to verify
Landing year 1976; landing sites (Chryse Planitia, Utopia Planitia)	C	NSSDCA Viking 1 and Viking 2 entries
Orbiter view shows Viking 1 back shell about 260 m (850 ft) away; "Big Joe" near Viking 1; "Ankylosaurus" near Viking 2	V	NASA MRO release (via NBC News and Mars Daily reports)
No organic molecules found at tested sites	C	NASA Viking science summary
Viking 1 last contact November 1982; cause often described as a faulty software command	C	NASA Viking 1 page. Include the cause only if NASA states it.

Developer notes

- The chapter must end on an open puzzle, not a win. This is intentional: it makes the next question feel earned.
- Do not say "there is no life on Mars". Say what the instruments found and did not find.

6.4 Chapter 4: Spirit and Opportunity

Banner in	**Was Mars once wet?**
Banner out (next chapter)	**Could anything fly on Mars?**
Status chip	SILENT [C]
Where it is	Mars. Gusev crater (Spirit), Meridiani Planum (Opportunity); landed January 2004 [V date, C places]
3D model file	ch4-mer-rover.glb (one model serves both rovers; they were twins [C])
"Where it is now" photo	ch4-hirise-opportunity-landing.jpg HiRISE color image: Opportunity's lander, parachute and heat shield are still where the rover left them [V]. Alternate: Spirit lander and Bonneville crater [V]. Credit: NASA/JPL-Caltech/UArizona. HiRISE images are public domain [V]

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Rock Abrasion Tool	A grinder that scrapes the dusty surface off a rock to show fresh rock beneath. [C]	Fresh rock tells a truer story than dusty rock.
2	Panoramic camera	The rover's eyes on a tall mast. [C]	Pictures chose which rocks to visit.
3	Solar panels	Sunlight is the power source. Dust on the panels is a problem. [C]	Power decides how long a rover lives. It is its strength and its weakness.
4	Wheels	Six wheels that climb over rocks and cross sand. [C]	Soft sand can still trap a rover. Spirit got stuck in 2009. [C]

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	Two rovers, Spirit and Opportunity, landed on opposite sides of Mars in January 2004 [V]. They were built to last about 90 days. [C]	FACT
2 How it works	Tap the glowing dots. A grinder exposes fresh rock. A camera chooses the next target. Solar panels feed the batteries. Six wheels do the driving.	FACT [C]
3 What we learned	Opportunity found minerals that usually form in water. Spirit found signs of past water too. Scientists inferred that parts of ancient Mars were wet. [C] Opportunity worked for over 14 years. [C]	INFERRED
4 Where it is now	Spirit went quiet in 2010 and Opportunity in 2018, after a giant dust storm blocked its sunlight. [C] Orbiter photos show the landing hardware is still there, though Mars wind has erased the tracks. [V]	FACT [V] [C]

Facts used and how to verify them

Claim	Tag	Where to verify
Both rovers landed January 2004	V	NBC News report on MRO images
Opportunity landing hardware still in place; wind has erased the tracks (HiRISE 2017 color image notes)	V	HiRISE image page "What Opportunity Left Behind"
Spirit: lander still bright, rover near Home Plate; no obvious tracks (wind erased them) per HiRISE image notes (2012)	V	HiRISE image page "Spirit Lander and Bonneville Crater in Color"
Built for about 90 sols; Opportunity lasted over 14 years; last contact June 2018; mission declared complete February 2019	C	NASA Opportunity mission page
Spirit stuck in soft sand 2009; last contact March 2010	C	NASA Spirit mission page
Water-formed minerals found by Opportunity and Spirit	C	NASA MER science results

Developer notes

- One IMAGINED line is allowed in this chapter, at step 4: "Imagine the quiet after the last signal." Label it IMAGINED.
- Be exact: "sols" are Mars days. If the word is used, explain it.

6.5 Chapter 5: Ingenuity helicopter

Banner in	**Could anything fly on Mars?**
Banner out (next chapter)	**Where does the Sun's reach end?**
Status chip	RETIRED [C]
Where it is	Mars. Jezero crater. Arrived with the Perseverance rover [C]
3D model file	ch5-ingenuity.glb (NASA 3D Resources if available [C])
"Where it is now" photo	ch5-ingenuity.jpg NASA Image and Video Library (search "Ingenuity"). Use an official NASA photo of the helicopter on the surface [C]

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Rotors	Two sets of blades spin in opposite directions, very fast, because Mars air is very thin. [C]	Engineers tested it in a huge airless chamber on Earth first. [C]
2	Solar panel	It charges the battery each day. [C]	The same solar-power strength and limit as the rovers.
3	Camera	It helps the helicopter see where it is flying. [C]	A flying camera sees farther than a rover can. [C]
4	Warm core	It keeps the electronics from freezing on cold nights. [C]	Surviving the night took as much cleverness as flying.

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	Ingenuity was a small helicopter that rode to Mars with a rover. It was a test: could anything fly in air this thin? [C]	FACT [C]
2 How it works	Tap the glowing dots. Fast blades grip thin air. A panel charges the battery. A camera guides the way. A warm core survives the night.	FACT [C]
3 What we learned	It made the first powered, controlled flights on another planet. It was planned for only a few flights and made 72. Engineers learned that aircraft can scout ahead of rovers. [C]	FACT [C]
4 Where it is now	Ingenuity's mission ended in January 2024 after its rotor blades were damaged. [C] It still sits on the surface of Mars.	FACT [C]

Facts used and how to verify them

Claim	Tag	Where to verify
First powered controlled flight on another planet; 72 flights; ended January 2024 (rotor damage)	C	NASA Ingenuity mission page
Rotor speed about 2,400 rpm	C	NASA Ingenuity technical page. Show only if confirmed.
Arrived with Perseverance in February 2021	C	NASA Perseverance mission page

Developer notes

- COULD: "Earth speed vs Mars speed" toggle on the rotors. Only add numbers that are confirmed [V].
- Ingenuity was a technology demonstration, not a science mission. Say so; it is a good lesson in how engineers prove ideas.

6.6 Chapter 6: Voyager 1

Banner in	**Where does the Sun's reach end?**
Banner out (next chapter)	**(banner goes blank)**
Status chip	STILL TALKING [V]
Where it is	Interstellar space. About 25.9 billion km from Earth at one light-day [V]
3D model file	ch6-voyager.glb (NASA 3D Resources; very likely available [C])
"Where it is now" photo	ch6-pale-blue-dot.jpg "Pale Blue Dot", Voyager 1's last photo, taken in 1990 from about 6 billion km [V]. Credit NASA/JPL-Caltech. Caption: "Too far to photograph. This is the last picture it ever took."

Hotspots (what the visitor sees when tapping a part)

#	Part	Its job (shown first)	What we learned because of it (shown second)
1	Big dish antenna	A dish that must stay pointed at Earth, or the connection is lost. [V]	It sends data at only about 160 bits per second, like very old dial-up. [V]
2	Nuclear power (RTGs)	Heat from radioactive material makes electricity. [V]	There is no sunlight out here, so power came from inside. It fades slowly, and runs out in the 2030s. [V]
3	Golden Record	A record of sounds and pictures from Earth, in case anyone ever finds it. [V]	It is mostly a symbol of who we are. [V]
4	Instrument boom	Sensors that measure particles and magnetic fields. [C]	They showed when the probe left the Sun's bubble. [C]

Step copy (draft, on screen)

Step	Copy	Label
1 Meet it	Voyager 1 launched in 1977 [V]. It is the farthest human-made object, and it is still talking to Earth. [V]	FACT [V]
2 How it works	Tap the glowing dots. A dish stays pointed home. Nuclear power runs the radio. A golden record carries a message from Earth.	FACT [V]
3 What we learned	It flew past Jupiter and Saturn, then kept going. In 2012 it became the first spacecraft to enter interstellar space. [C] Now a radio message takes about a day to reach it. [V]	FACT [V] [C]
4 Where it is now	LIVE PANEL (see below). Final line: "A command sent on Monday morning gets an answer on Wednesday morning." [V]	FACT [V]

Facts used and how to verify them

Claim	Tag	Where to verify
One light-day from Earth: about 10:16:07 UTC on Nov 18, 2026 (about 25.9 billion km); NASA notes the exact moment can shift slightly	V	NASA announcement as reported by IFLScience and EarthSky, June 2026. Re-check on the NASA Voyager page.
Speed about 38,000 mph (about 17 km per second)	V	JPL Voyager project manager quote (CNN / ABC17 report)
Data rate about 160 bits per second	V	JPL project manager quote (Jang report). Prefer a NASA page if available.
Voyager 2 reaches one light-day around November 2035	V	CNN / ABC17 report
Last photo "Pale Blue Dot", February 14, 1990	V	IFLScience; NASA image library
Entered interstellar space in 2012	C	NASA Voyager page

Developer notes

- The live panel is described in section 8.1. It must work offline with the fallback formula.
- The chapter ends without a "Next question" machine. The Next button reads "Add your question" and starts the finale.
- After Nov 18, 2026 10:16 UTC the countdown flips to: "Voyager 1 is now more than one light-day away."

THE ENDING BELONGS TO THE VISITOR

7 Finale and the Constellation card

After chapter 6, the banner erases slowly and stays blank with a blinking cursor. The narrator says: **"Voyager is still going. Your turn."** The visitor writes the next question. The six chapter questions and the visitor's question become one constellation that they can keep.



Figure 7.1 The Constellation. Six chapter stars joined in order; the last, larger star is the visitor's own question (dashed link).

Step	Specification
F1 Blank banner	Banner text erases to empty over 1200 ms. Cursor blinks (static cursor with reduced motion). Narrator line fades in under the banner.
F2 Write	Text field labeled "What do you want to know next?". Placeholder: "I wonder if...". Counter: 140 characters max, minimum 3. Three tap-to-fill examples: "Is anyone else out there?", "What is under the ground on Mars?", "How far can a machine go?". Enter or "Add my star" submits.
F3 Check	Trim spaces. Reject links, email addresses and phone numbers (regex) and a short list of unsafe words, with a friendly message: "Keep it a question about space. No names or contact details, please." Nothing is sent to any server.
F4 Build	Seven stars appear one by one, 350 ms apart; connecting lines draw in 500 ms each; the visitor's star arrives last, larger, with a gold ring. With reduced motion: appear all at once.
F5 Card	Card preview, 4:5 ratio, exported at 1080 x 1350 px PNG. Layout: top line "MY QUESTION FOR THE NEXT MACHINE" (small caps, Moon Silver); the question in Question Gold, centered, up to 5 lines; the constellation in the lower half; footer "Built for a Question | NASA Space Apps Challenge 2026 | Not an official NASA product". Starfield is seeded by a hash of the question so each card is unique and repeatable.
F6 Actions	Buttons: "Download card" (PNG), "Share" (Web Share API when available; otherwise hidden), "Start again" (clears the question, returns to #ch1). Final line below: "Nothing out there was abandoned. It was handed on. What will you ask next?"

Children's privacy (MUST)
No accounts, no cookies, no analytics that identify a person, no server storage of the question, no external trackers or ad scripts. The card is drawn on a canvas and saved on the visitor's own device. State this in one sentence on the finale screen: "Your question stays on your device."

REAL DATA, SAFE ANSWERS

8 Live data and the chat box

8.1 Voyager 1 live panel (Chapter 6, step 4)

The panel shows four numbers that update every second. It must keep working with no network.

Field	Formula or source	Display
Distance from Earth	See fallback below, or the precomputed table	"25.9 billion km" (1 decimal)
One-way signal time	distance in km divided by 299,792.458 (km per second)	"23 h 58 min" (hours and minutes)
Message and reply	2 times the one-way time	"about 2 days"
Speed	about 17 km per second (38,000 mph) [V]	"about 17 km every second"
Countdown to one light-day	Target: 2026-11-18T10:16:07Z [V]. Show days, hours, minutes.	After the target: "now more than one light-day away"

Recommended data strategy (most robust):

- **Before the event:** fetch a daily table of Earth to Voyager 1 distance for Nov 1 to Dec 31, 2026 from JPL Horizons (target ID -31 for Voyager 1 [C], observer Earth center) and save it as `/data/voyager1.json`. At runtime, interpolate between rows.
- **Fallback if the table is missing:** distance(t) = 25,902,068,371 km + 17.0 km/s x (t minus 2026-11-18T10:16:07Z in seconds). One light-day is 299,792.458 x 86,400 = 25,902,068,371 km. This fallback is only accurate to about 0.6 percent because Earth's own motion changes the distance. Show it with "about".
- **Optional live call** to the Horizons API from the browser, if CORS allows [C]. Always keep the table and the fallback.

> **Accuracy line to show under the panel**
> "Distance is calculated from NASA/JPL tracking data. NASA says the exact moment Voyager 1 reaches one light-day may shift slightly."

8.2 Ask the engineer (chat)

A small chat panel that answers children's questions about the machines. It exists to feel friendly and helpful, but its first duty is to be correct. Two build options:

Option	How it works	Risk	Recommendation
A: Retrieval only (no AI model)	Search the fact cards with a small library (Fuse.js or MiniSearch). Return the best card's "short answer" text and its source link. A set of 12 starter questions as chips.	Very low. Answers can only be text we wrote.	**MUST build this**
B: Retrieval + language model	Same search, then a language model rewrites the answer in simple words, using only the retrieved cards.	Medium. A model may add outside claims.	COULD, only after A works and with the prompt below

Rule	Specification
Scope	Answers only from fact cards. If nothing matches: "I do not have that in my notes. Here is where scientists look:" followed by 1 or 2 source links.
Reading level	Ages 9 to 14. Maximum 3 sentences. Include the source link.
Label	Every reply shows the Truth Label of the card it used and a footer: "Dramatized voice, facts from NASA."
Off-topic and unsafe	Polite refusal: "I can only talk about the machines in this story." No personal chat, no advice, no opinions.
Never	Never invent numbers, dates or quotes. Never say a machine "said" something unless the fact card says so. Never claim there is or is not life on Mars.
Privacy	Do not log or store questions. No third-party calls in Option A.

System prompt for Option B (copy exactly):

```
You are "the engineer", a friendly guide for children aged 9 to 14 on a NASA Space Apps
student website. Answer ONLY using the FACT CARDS provided in this message.
If the cards do not contain the answer, say: "I do not have that in my notes."
Never add numbers, dates, names or quotes that are not in the cards.
Use at most 3 short sentences. End with the card title in brackets.
Never claim that machines speak, feel or think. Never discuss anything off-topic.
```

Test questions the chat must pass (use them in QA):

Question	Expected behavior
Why did Opportunity go quiet?	Answer from Ch4 card: dust storm blocked sunlight in 2018 [C]. Cites source.
How does Ingenuity fly?	Answer from Ch5 card: fast opposite-spinning rotors for thin air.
Is there life on Mars?	Answer from Ch3 card: what the Viking labs found and did not find. Never says yes or no.
How far is Voyager 1?	Reads the live panel value and the light-time.
Who will win the World Cup?	Polite refusal (off-topic).
What is the best rover?	Refuses to rank. Offers to compare what two machines were built to do.
Tell me about the Mars Pathfinder rover.	"I do not have that in my notes." + source links (it is not in the six chapters).

8.3 Fact card format

Every chapter is rendered from one JSON file. Writers edit JSON; developers never hard-code copy in components. The same file feeds the chat.

```
{
  "id": "ch4-mer",
  "title": "Spirit and Opportunity",
  "chapter": 4,
  "bannerIn": "Was Mars once wet?",
  "bannerOut": "Could anything fly on Mars?",
  "status": { "chip": "SILENT", "note": "Opportunity: last contact June 2018 [C]" },
  "model": "/models/ch4-mer-rover.glb",
  "modelCredit": "NASA/JPL-Caltech (confirm)",
  "hotspots": [
    { "id": "rat", "label": "Rock Abrasion Tool", "orbit": "40deg 70deg 1.1m",
      "job": "A grinder that scrapes ...", "learned": "Fresh rock tells ..." }
  ],
  "steps": [
    { "name": "Meet it", "text": "Two rovers ...", "label": "FACT" }
  ],
  "photo": { "src": "/img/ch4-hirise.webp", "alt": "...", "caption": "...",
             "credit": "NASA/JPL-Caltech/UArizona", "sourceUrl": "https://..." },
  "facts": [
    { "claim": "Both rovers landed Jan 2004", "tag": "V", "source": "https://...",
      "checkedOn": "2026-11-08", "checkedBy": "AB, CD" }
  ],
  "shortAnswer": "Opportunity went quiet in 2018 after a dust storm. [C]"
}
```

HOW IT LOOKS AND FEELS

9 Visual design system

Mood: a quiet observatory at night. Dark, calm, a little emotional. Gold is reserved for questions. Teal is reserved for things that are alive or tappable. Real photos and 3D are the stars; interface chrome stays minimal.

Deep Space
#0B1020
page background

Midnight Panel
#141B34
cards, panels

Panel Raised
#1E2748
hover, raised panels

Starlight
#F4F1EA
main text

Moon Silver
#B8C0CC
secondary text

Question Gold
#FFC857
Question Banner, main CTA

Signal Teal
#3DD6C6
hotspots, live data, FACT

Mars Rust
#E4572E
Mars, SILENT status

Dust Violet
#8A7CFF
IMAGINED label

Amber
#F5A524
INFERRED label

Figure 9.1 Color palette with hex codes. Use only these colors.

Pair (text on background)	Approx. contrast	Use
Starlight on Deep Space	about 16 : 1	Body text
Moon Silver on Deep Space	about 10 : 1	Secondary text, captions
Question Gold on Deep Space	about 13 : 1	Banner, buttons (use Deep Space text on gold fills)
Signal Teal on Deep Space	about 10 : 1	Hotspots, live numbers
Mars Rust on Deep Space	about 5 : 1	Large text and icons only
Dust Violet on Deep Space	about 5.5 : 1	IMAGINED label

Contrast values are estimates. Developers must verify every pair with a contrast checker and meet WCAG 2.2 AA (4.5 : 1 for body text, 3 : 1 for large text and UI shapes).

9.2 Typography

Role	Font (self-host, all open license [C])	Size and rules
Headings and banner	Space Grotesk, Bold	Banner 22 to 30 px; chapter titles 32 to 44 px. Line height 1.15.
Body	Atkinson Hyperlegible (designed for legibility)	Minimum 18 px; line height 1.6; max line length 65 characters.
Numbers and live data	JetBrains Mono	Tabular figures so digits do not jump as they change.
Fallbacks	system-ui, sans-serif	Always provide; use font-display: swap.

9.3 Components

FACT INFERRED IMAGINED STILL TALKING STILL ANSWERING SILENT RETIRED

First three: Truth Labels (on text). Last four: Status Chips (on each machine). STILL ANSWERING = a passive device that replies when contacted.

- **Buttons:** primary = Question Gold fill with Deep Space text, 44 px minimum height (touch target). Secondary = outline in Moon Silver.
- **Hotspot ring:** 28 px teal ring with a 10 px filled center; 44 px invisible tap area.
- **Panels:** Midnight Panel background, 12 px radius, 1 px border in Panel Raised. No heavy shadows.
- **Icons:** simple outline icons, 2 px stroke (Lucide or Phosphor, MIT license [C]).

- **Photos:** 4:3 crop, 12 px radius, thin silver border. Always show the credit line in 12 px Moon Silver under the photo.
- **3D lighting:** one soft key light, one cool rim light, no HDR background; background is a dark radial gradient with a faint starfield.
- **Sound:** none by default. Optional ambient tone and the seismometer rumble are opt-in with captions.

9.4 Design tokens (copy into the project)

```
:root {
  --deep-space:#0B1020; --panel:#141B34; --panel-raised:#1E2748;
  --starlight:#F4F1EA;  --silver:#B8C0CC;
  --gold:#FFC857; --teal:#3DD6C6; --rust:#E4572E; --violet:#8A7CFF; --amber:#F5A524;
  --font-head:"Space Grotesk",system-ui,sans-serif;
  --font-body:"Atkinson Hyperlegible",system-ui,sans-serif;
  --font-mono:"JetBrains Mono",ui-monospace,monospace;
  --radius:12px; --space:8px; --tap:44px;
}
@media (prefers-reduced-motion: reduce) { * { animation:none !important; transition:none !important; } }
```

HOW TO BUILD IT

10 Technical specification

Area	Recommendation	Notes
Framework	Vite + vanilla JavaScript or React. Static build, no server.	Pick what the team knows best. No server keeps hosting and privacy simple.
3D	Google `model-viewer` web component (self-hosted from npm, pinned version).	Supports hotspots, camera-orbit transitions and a poster fallback. Alternative: three.js with GLTFLoader.
Scroll	CSS position: sticky + IntersectionObserver. Optional GSAP ScrollTrigger (COULD).	No wheel hijacking.
Search for chat	Fuse.js or MiniSearch over the fact-card JSON.	Runs in the browser.
Card export	HTML canvas, then canvas.toBlob to PNG.	No server upload.
Hosting	Netlify, Vercel or GitHub Pages (static).	Use a clear project URL for the submission.
Browsers	Last 2 versions of Chrome, Edge, Safari, Firefox. Test on a mid-range Android phone.	Many users will be on mobile data.
Languages	English first. Bangla (BN) as a COULD: all copy lives in JSON so it can be translated.	Translations must be checked by a fluent speaker.

10.1 Data flow

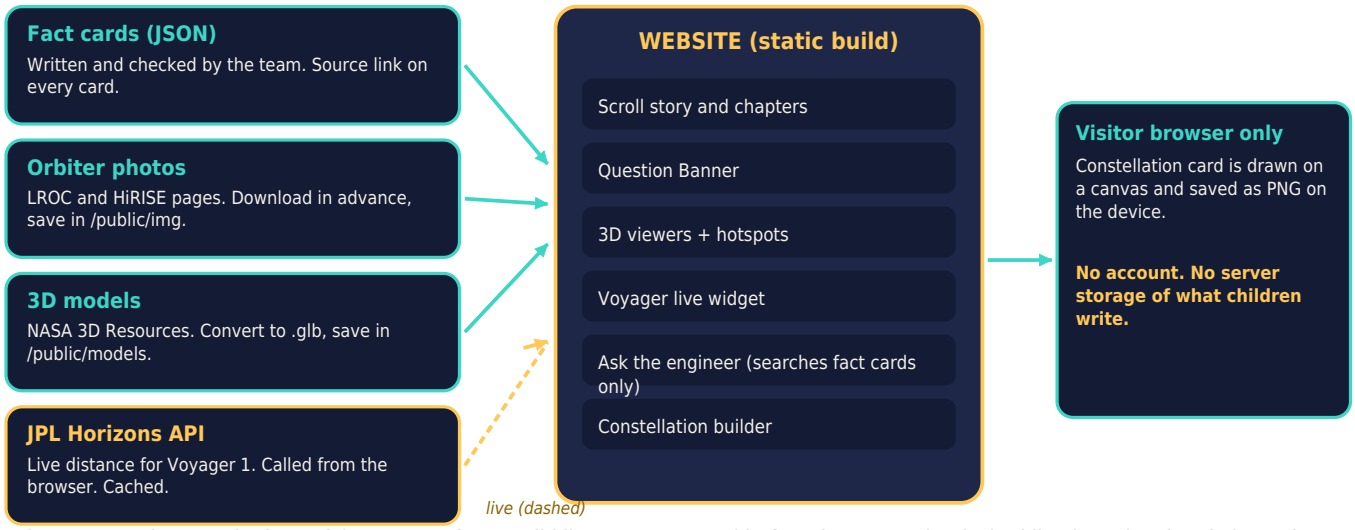


Figure 10.1 Where each piece of data comes from. Solid lines are prepared before the event; the dashed line is optional and always has a fallback.

10.2 Project structure

```
/public
  /models   ch1-retroreflector.glb ... ch6-voyager.glb    (each 3 MB or less, Draco/meshopt)
  /img      ch1-lroc-apollo15.webp ... ch6-pale-blue-dot.webp  (each 250 KB or less)
  /fonts    SpaceGrotesk, AtkinsonHyperlegible, JetBrainsMono (woff2)
/data       ch1.json ... ch6.json   voyager1.json   chat-starters.json   sources.json
/src        main.js   banner.js   chapter.js   viewer.js   voyager.js
            chat.js   finale.js   settings.js   a11y.js     styles.css
/docs       README.md   FACT-CHECK-LOG.md   CREDITS.md
```

10.3 Performance budget

Metric	Target
First load (excluding 3D)	Under 2.5 MB total; JavaScript under 250 KB gzipped
Largest Contentful Paint	Under 2.5 s on a mid-range phone over 4G
3D models	Load chapter N+1's model when chapter N reaches step 3. Each model under 3 MB. Show the poster image while loading.
Layout shift	Under 0.1 (reserve space for images and the scene)
Frame rate	30 fps or better on a mid-range phone; drop auto-rotate if the frame rate falls

10.4 Accessibility checklist (target WCAG 2.2 AA)

Requirement	How we meet it	Priority
Keyboard only	Every control reachable and visible focus ring (3 px gold). Hotspots are buttons. Esc closes part cards.	MUST
Screen readers	Banner is aria-live polite. Each 3D scene has a text alternative describing the machine and listing the parts. Photos have alt text.	MUST
Reduced motion	Follows the OS setting and a manual toggle. See the table in 5.5.	MUST
Text size and contrast	Three text sizes, high-contrast mode, no information by color alone (labels carry text).	MUST
No WebGL	If 3D fails or is slow, show the poster PNG and the same hotspot list as buttons.	MUST
Read aloud	Web Speech API button on the story panel (hide if unsupported).	SHOULD
Captions	Any sound has captions. Video has burned-in subtitles.	MUST
Seizure safety	No flashes above 3 per second.	MUST
Plain language	Reading level checked per chapter.	SHOULD

10.5 Test matrix

Device	What to test
Mid-range Android phone, Chrome	Scroll feel, bottom sheet, 3D speed, finale card download
iPhone, Safari	Sticky sections, model-viewer, share sheet, audio rules
Laptop, Chrome and Firefox	Full flow, keyboard only, screen reader (NVDA or VoiceOver) for one chapter
Throttled network (Slow 4G) and offline	Poster fallbacks, Voyager fallback formula, no blank screens

HOW WE FINISH

11 Plan, roles, scope, risks and QA



Prep covers research, writing, asset collection and planning. Confirm in the Nov 13 guide what may be built before the event.

Figure 11.1 Plan from today (Oct 30) to Voyager 1's one-light-day moment (Nov 18). The hackathon is Nov 14 and 15; submissions close Nov 15 at 11:59 PM local time [V].

11.1 Roles (one person can hold several)

Role	Owns	Done when
Lead and integrator	Repo, merges, deadlines, final submission, rule check	Submission page complete before the cutoff

Role	Owns	Done when
Front-end / 3D	Chapter template, viewer, hotspots, camera states, banner animation	All 6 chapters run from JSON
Content and fact-checker	Fact cards, [C] to [V] upgrades, reading level, Truth Labels, credits	Every [C] claim is verified or cut
Designer	Tokens, wireframes to final, photo crops, finale card, icons	Design matches section 9
Video and presentation	Storyboard, recording, captions, project summary text	Video exported and uploaded

11.2 Scope: what to build first

Priority	Items
MUST	Landing, Question Banner with rewrite, chapter template with 4 steps, 6 chapters filled from JSON, hotspots, photos with credits, finale with card download, Truth Labels, Sources drawer, reduced motion, keyboard access, privacy notice, footer disclaimer
SHOULD	Voyager live panel with fallback, Distance Ruler, Ask the engineer (Option A), read aloud, "Fire laser" mini-interaction in chapter 1, hand-off travel animation
COULD	Bangla language, Option B chat, tap-to-define words, seismometer rumble, rotor speed toggle, a seventh machine (for example GRAIL or Phoenix)

Cut line
If by the middle of day 2 the MUST list is not green, drop all SHOULD and COULD items and polish the MUST list. A finished small project beats an unfinished big one.

11.3 Hackathon schedule (adjust to your local hours)

Block	Goal
Day 1, first half	Chapter template working with chapter 1 from JSON: banner, steps, one hotspot, photo. Merge.
Day 1, second half	Chapters 2 to 6 filled in. Finale and card. Voyager panel with fallback.
Day 2, first half	Accessibility pass, mobile pass, performance pass, chat Option A, sources drawer.
Day 2, second half	Freeze features. Record video. Write the submission text. Test the final link on a phone. Submit with time to spare.

11.4 Definition of done

- All six chapters complete all four steps with correct copy from the fact cards.
- Every claim on screen is [V] with a source, or has been cut. The FACT-CHECK-LOG shows two sets of initials per claim.
- The banner rewrites correctly between every pair of chapters, including with reduced motion.
- The site works on a mid-range phone and with the keyboard only.
- If 3D fails, the page still tells the whole story with photos.
- The finale card downloads on desktop and phone and shows the visitor's question.
- No network calls other than the optional Horizons request. No trackers.
- Credits page lists every photo, model, font and library with its license.

11.5 Risk register

Risk	Likelihood	Mitigation
A fact is wrong	Medium	Two-person fact check; [C] claims cannot ship; chat limited to fact cards
A 3D model is not available or too large	Medium	Build a simple labeled illustration; compress with Draco; poster fallback

Risk	Likelihood	Mitigation
Photos have unclear usage rights	Medium	Use NASA and HiRISE images with credit; read each page's usage note; keep a CREDITS.md
Live data call fails during the demo	Medium	Use the precomputed table and the fallback formula; record the video with data visible
Scope too big for the weekend	High	Template-first build; cut line in 11.2; six chapters only
Phone performance is poor	Medium	Lazy-load models; turn off auto-rotate; posters first
Pre-event work breaks a rule	Unknown	Read the 2026 submission guide on Nov 13 and do only what it allows before the event

11.6 Decisions the team lead must make this week

- Names for each role and how many people are on the team.
- Framework (Vite + React or vanilla) and 3D approach (model-viewer or three.js).
- Whether Bangla is in scope.
- Hosting service and the project URL.
- Who confirms every [C] claim, and by which date.

WHERE FACTS AND ASSETS COME FROM

12 Sources and credits

Rows tagged [V] were opened and read while preparing this brief. Rows tagged [C] are standard NASA resources named in the challenge or well known, but their exact page addresses must be confirmed by whoever uses them.

Source	Use	Tag
spaceappschallenge.org/2026 (challenge page for "Abandoned but not Forgotten")	Official brief and resources	V
spaceappschallenge.org/resources/judging-awards-guide	Judging criteria (earlier years); re-check Nov 13	V
uahirise.org/ESP_050177_1780 (What Opportunity Left Behind)	Opportunity landing hardware; tracks erased by wind	V
hirise.lpl.arizona.edu/ESP_025815_1655 (Spirit Lander and Bonneville Crater)	Spirit lander view	V
news.asu.edu (New LROC images offer sharper views of Apollo 12, 14, 17 sites)	Apollo site photos	V
space.com/12835 (NASA unveils new detailed photos of Apollo landing sites)	Apollo footpaths and rover tracks seen from orbit	V
nbcnews.com/id/wbna16055585 (MRO images of Viking and rover sites)	Viking back shell, "Big Joe", Spirit and Opportunity landing year	V
iflscience.com (Voyager 1 will cross the one light-day threshold on Nov 18, 2026)	Target date and time; Pale Blue Dot	V
earthsky.org (Voyager 1, 1 light-day from Earth, Nov 17/18, 2026)	Light-day date	V
kvia.com (CNN: Voyager 1 will reach one light-day in 2026)	Speed, one-day signal delay, Voyager 2 in 2035	V
spacedata.swapniltamse.com (Space Data Atlas: challenge resources)	List of challenge datasets (see below)	V
NASA Image and Video Library, NASA 3D Resources, JPL Horizons, NSSDCA Master Catalog, Planetary Data System, Apollo Lunar Surface Journal	Photos, models, distance table, mission facts, ALSEP data, astronaut words	C

Datasets named for this challenge (as listed on the Space Data Atlas page) [V]: NASA Catalogue of Manmade Material on the Moon (796 items), LROC Image Archive and Featured Sites, Apollo ALSEP data recovered from archival tapes (PDS), HiRISE images from the Mars Reconnaissance Orbiter, the NSSDCA Master Catalog, and the Apollo Lunar Surface Journal.

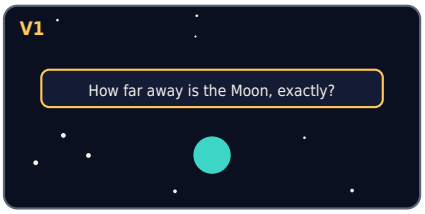
Credit lines
Every photo must show its credit exactly as the source gives it (for example "NASA/JPL-Caltech/UArizona", "NASA/GSFC/Arizona State University"). Every library and font goes in CREDITS.md with its license. Add on the Sources page: "This project uses NASA data and images. NASA does not endorse this project."

HOW WE SHOW IT

13 Presentation video

The video has one job: let a judge understand the idea in the first 10 seconds and remember four moments afterward. The four moments are: **(1)** the banner rewriting itself, **(2)** tapping a part of a machine to see its job and its lesson, **(3)** the live Voyager 1 distance, and **(4)** the Constellation card with a child's own question.

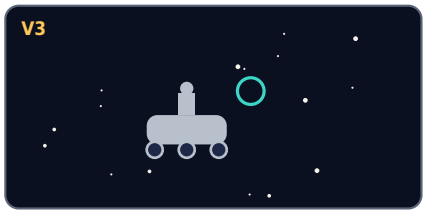
Length: plan a master of about 2.5 minutes and a 30-second cut. **The allowed length and format come from the 2026 submission guide (released Nov 13). Follow it exactly.** The timings below are for the video only, not for the website.



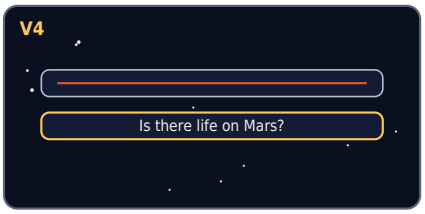
Hook



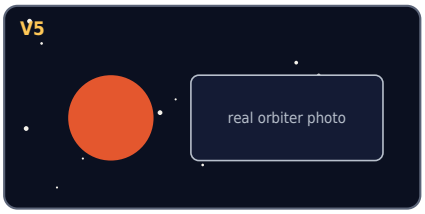
The idea



Demo: a machine



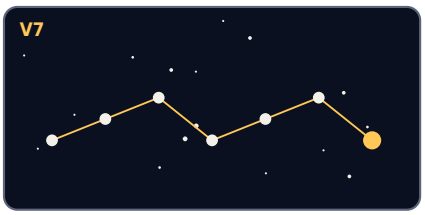
Demo: the hand-off



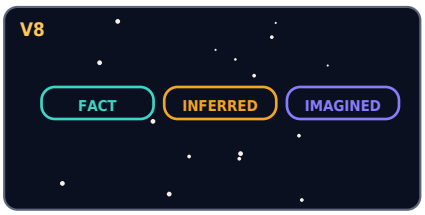
Demo: where it is now



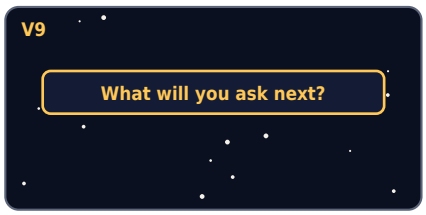
Live Voyager



Finale



Trust and access



Close

Figure 13.1 Storyboard frames V1 to V9.

13.1 Script and capture list

Scene	Time	What is on screen (record this)	Voice-over
V1 Hook	0:00-0:15	Black screen. The banner types: "How far away is the Moon, exactly?" Small Earth glows.	Out there, on the Moon, on Mars and beyond, machines sit quiet. Most people call them abandoned. We think they were handed on.
V2 The idea	0:15-0:35	Animated chain of six machine cards joining with arrows (Figure 2.1), then the six questions in gold.	Every machine NASA left behind was built to answer one question. And every answer created the next one. We turned that chain into a journey.
V3 A machine	0:35-1:00	Chapter 1. Tap the mirror hotspot. Press "Fire laser": the pulse goes and returns with the stopwatch.	Start at the Moon. These mirrors have no electronics, so nothing can break. Fire a laser from Earth, and the light comes back in about two and a half seconds. That is how we learned the Moon is drifting away.
V4 The hand-off	1:00-1:15	Step 4 to Next. The banner erases and types "Is there life on Mars?" while the camera pulls back and flies to Mars.	Then the question changes. And so does the world.
V5 Where it is now	1:15-1:35	Chapter 4. Tap the rock grinder. Crossfade to the real orbiter photo of Opportunity's landing site, with its credit line.	Opportunity was built for ninety days and worked for fourteen years. It went quiet in 2018. But NASA's orbiters still photograph its landing site. Nothing here is forgotten.
V6 Live Voyager	1:35-1:55	Chapter 6. The live panel: distance, one-way signal time, countdown.	Voyager 1 is still talking. On November eighteenth it reaches one light-day from Earth. A message takes a full day to arrive.
V7 Finale	1:55-2:15	The banner goes blank. Type a child's question. Stars connect. Press Download.	At the end, the banner goes blank. The last question belongs to you, and it becomes the last star.
V8 Trust and access	2:15-2:30	Quick shots: FACT / INFERRED / IMAGINED labels, the Sources drawer, Ask the engineer refusing an off-topic question, the Reduce motion toggle.	Every line is labeled fact, inferred or imagined. The chat only answers from sourced fact cards. And it works with a keyboard, reduced motion and read aloud.
V9 Close	2:30-2:40	Banner: "What will you ask next?" Team name and "Not an official NASA product."	Nothing out there was abandoned. It was handed on. What will you ask next?

Rule for the voice-over
Only say facts that are tagged [V] by the time you record. Replace any [C] claim in this script with a verified version, or cut the sentence.

13.2 The 30-second cut

Seconds	Shot
0-5	V1: the banner types its question over a dark Earth.
5-11	V4: banner rewrites; camera flies to Mars.
11-19	V6: live Voyager 1 distance and countdown.
19-25	V7: a child's question becomes the last star; card downloads.
25-30	V9: "Nothing out there was abandoned. It was handed on."

13.3 Recording checklist

- Record the screen at 1920 x 1080, 60 fps, with a clean browser profile, no bookmarks bar, and the Voyager panel online.
- Record the voice-over in a quiet room with a headset or phone microphone close to the mouth; remove background noise; keep the same level throughout.
- Burn in English subtitles; add Bangla subtitles if the team has a reviewer (COULD).
- Music: royalty-free, quiet, never louder than the voice. Credit it.

- Show every NASA credit line on screen when its image appears.
- Export H.264 MP4. Check the length and file-size limits in the 2026 guide, then upload and test the link while logged out.

13.4 Draft project summary for the submission form (about 100 words)

DRAFT

"Built for a Question" is a scroll-driven story website for ages 9 to 14. Every machine NASA left on the Moon, on Mars and in deep space was built to answer one question, and each answer created the next. Visitors follow that chain from Apollo's laser mirrors to Voyager 1, tapping 3D parts to see how the engineering produced the science, and see real orbiter photos of where each machine sits today. A live Voyager 1 distance counter, Truth Labels (fact, inferred, imagined) and a sourced fact-card chat keep it accurate. The ending asks the visitor to write the next question and download it as a constellation card.

13.5 Questions judges may ask, and our answers

Question	Answer
Isn't this just a museum of old machines?	A museum lets you browse. This is a relay: the order and the question chain are the story, and every chapter shows how a design choice produced a discovery.
Why not a game?	The brief asks us to tell a story and show how missions operated. Our interactions (tap a part, fire a laser, write a question) serve the story instead of competing with it.
How do you keep it accurate?	Source-linked fact cards, two-person checks, Truth Labels on every block, and a chat that can only answer from the cards.
What real data do you use?	NASA orbiter photos (LROC, HiRISE), NASA mission records, Apollo instrument data, and live Voyager 1 distance from NASA/JPL tracking.
Who is it for?	Ages 9 to 14, plus teachers and parents. Keyboard, reduced motion, large text and read aloud are built in.
What if 3D or the network fails?	Poster images tell the story, and the Voyager panel falls back to a calculated distance.
What would you build next?	More machines (GRAIL, Phoenix, InSight, Pioneer), Bangla language, and a teacher guide with classroom questions.

FIXED STRINGS (PUT THESE IN JSON)

B UI copy deck

Where	Text
Landing headline	Nothing out there was abandoned.
Landing subline	NASA left machines on the Moon, on Mars and in deep space. Each was built to answer one question. Follow the questions.
Landing button	Begin
Landing small print	A student project for the NASA Space Apps Challenge. Not an official NASA website.
Scroll prompt	Scroll to leave Earth
Hotspot hint (step 2)	Tap the glowing dots
Next buttons	Next | Next question | Add your question (chapter 6)
Back button	Back
Status chips	STILL TALKING | STILL ANSWERING | SILENT | RETIRED
Truth Labels	FACT | INFERRED | IMAGINED
Truth Label legend	FACT: measured and sourced. INFERRED: scientists' best explanation. IMAGINED: a story to help you feel it.
Footer (every page)	Dramatized voice, facts from NASA. Not an official NASA website; not endorsed by NASA.

Where	Text
Sources button	Sources
Chat button (aria-label)	Ask the engineer
Chat placeholder	Ask about a machine...
Chat no-match	I do not have that in my notes. Here is where scientists look:
Chat off-topic	I can only talk about the machines in this story.
Finale label	What do you want to know next?
Finale privacy line	Your question stays on your device.
Finale validation	Keep it a question about space. No names or contact details, please.
Finale buttons	Add my star | Download card | Share | Start again
Card title	MY QUESTION FOR THE NEXT MACHINE
Card footer	Built for a Question | NASA Space Apps Challenge 2026 | Not an official NASA product
Final line	Nothing out there was abandoned. It was handed on. What will you ask next?
3D fallback note	The 3D view could not load. Here is the picture and the list of parts instead.
Voyager after target time	Voyager 1 is now more than one light-day away.

End of brief

Questions about the story, the facts or the priorities go to the lead and the content owner before code is written, not after. Keep this PDF as the reference; update the version number and date on the cover when anything in a LOCKED DECISION changes.